

USTECH
INNOVATIONS

USTECH MasterLine17EC
Feed Quality Assured in Under a Minute



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Designed for ultimate ease of use, the USTECH MasterLine delivers highly accurate feed and forage analysis in under a minute. It empowers manufacturers to confidently control quality, optimize production costs, and maintain strict product consistency with fast, direct, and reliable measurements.







Dynamic & Removable Sample Table

The system features a specially designed rotating sample table that ensures a fully representative scan of your entire material batch. Beyond its rotational precision, the table is easily removable, allowing for effortless cleaning between tests. This guarantees zero cross-contamination and unmatched practical efficiency in your daily laboratory workflow.

High-Precision Optical Sensor

Strategically positioned beneath the rotating table, the advanced optical sensor continuously scans every point of the passing material. It detects the spectral responses of critical parameters—such as fat, protein, starch, and crude fiber —in mere milliseconds. This flawless data flow guarantees consistent, lab-grade accuracy with every single measurement.

Fast & Non-Destructive Analysis

Designed for the analysis of a wide range of feed and agricultural materials, USTECH MasterLine supports fast, non-destructive measurements of raw materials, compound feeds, forage, and grains. Its flexible sample handling capabilities allow both ground and unground materials to be analyzed, simplifying routine quality control and supporting efficient laboratory workflows.



Technical Specifications

Measurement Mode	Solid/Liquid
Wavelength Range	950 – 1650 nm (10,526 – 6060 cm-1)
Spectral Bandwidth	<1.25% of center wavelength (1% typical)
Detector	InGaAs photodiode array
Dynamic range (Max)	1000:1
Integration time	10 ms typical, minimum 10 µsec
Signal-to-noise ratio (typical)	23,000 (average of 100 scans)
Number of scan	2000 spectra/sample
Bulb life	>40,000 hr
Reference	Spectralon (>99% Diffuse Reflectance)
Absorbance range	Up to 3 AU
Sample Type	Direct measurement (ground/unground/ pellets)
Parameters	Fat, protein, moisture, ash, starch and crude fiber

***Adjustable

Industry Applications & Workflows



Raw Material Processing

Effortlessly analyze incoming raw materials like wheat, maize, soybean meal, and sunflower meal . This material-specific calibration ensures that out-of-spec ingredients are prevented from entering the production line, securing quality from the very first step.



Poultry & Ruminant Feed Production

Instantly verify the exact nutritional value of chicken feed and ruminant feed formulations . Guarantees that the final product meets the precise dietary and metabolic requirements necessary for optimal animal growth and health.

Installation Requirements

Function Range	0 - (+)35 °C**
Storage/Transport Range	-20 - (+)40°C**
Mechanical Environment	Stationary
EMC Environment	Laboratory Use
USB Connector	USB 2.0 Type A to B
Power Supply	2.1 Plug-in
Input Voltage	176-264 V
Frequency	50-60 Hz
Output Voltage	24 V DC
Output Current	3.6 A
Dimentions	250x355x170 mm
Housing	Aluminum

**Specifications may be subjected to change without prior notice
**Humidity ≤ 85% at 35°C

Our Core Commitments to You

"The Standard in Spectroscopy" is our commitment to your success. We provide cutting-edge analytical solutions so you can focus entirely on optimizing production, streamlining workflows, and ensuring uncompromised product quality.



Innovative

We leverage advanced technological expertise to develop state-of-the-art solutions—like direct measurement systems—that redefine laboratory standards.



Reliable

We guarantee precision and reproducibility, ensuring consistent, lab-grade results you can trust in every analysis.



Effortless

We simplify your daily workflow with intuitively designed instruments requiring zero-to-minimal sample preparation.



Cost-Effective

Maximize economic benefit by completely eliminating the need for expensive consumables, toxic reagents, and time-consuming methods.



Safe & Clean

We prioritize operator safety and the environment through robust housing architectures and chemical-free analytical processes.



Fast

Empowering rapid decisions with instant insights. We deliver accurate results in under a minute to keep your production moving.

CONTACT US

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